Termination of flips

Based on a Lecture series by Nikolaos Tsakanikas.

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We will discuss the 'termination of flips conjecture' in birational geometry, a fundamental conjecture of the minimal model programme.

§ 0.1 About. This course on ??? was a part of a Winter School for students and early career researchers, organised by the UK Algebraic Geometry Network. The lectures were delivered in-person in the University of Lancaster, and were transcribed by Parth Shimpi. These notes have undergone several amendments and are not a verbatim recall of the lectures, therefore discretion is advised when using this material. They are available online at `https://pas201.user.srcf.net/documents/2024-ukag-flips.pdf`. All errors and corrections should be communicated to by email to `parth.shimpi@glasgow.ac.uk`.

§ 1 Birational equivalence

Recall two varieties X, Y are said to be *birational* or *birationally equivalent* if any of the following equivalent conditions hold.

1. There exists a birational map $X \dashrightarrow Y$, i.e. a dominant rational map $X \dashrightarrow Y$ with a dominant rational inverse $Y \dashrightarrow X$.
2. There exist dense open subsets $U \subseteq X$, $V \subseteq Y$ such that U is isomorphic to V.
3. The varieties X, Y have isomorphic function fields, i.e. there is an isomorphism of $\mathbb{C}$–algebras $\mathbb{C}(X) \cong \mathbb{C}(Y)$.

Example 1.1 (A cuspidal cubic curve). Consider $C = \mathbb{V}(y^3 - x^2)$ as a subset of $\mathbb{A}^2$. The map $f : \mathbb{A}^1 \to C$ given by $t \mapsto (t^3, t^2)$ is a birational morphism which restricts to an isomorphism $\mathbb{A}^1 \setminus \{0\} \xrightarrow{\sim} C \setminus \{(0,0)\}$

Example 1.2 (Blowing up $\mathbb{A}^2$ at the origin). The space $X \subset \mathbb{A}^2 \times \mathbb{P}^1$ consisting of all points $(x, y, [u : v])$ such that $xv - yu = 0$, called the *total space of the blowup*, admits a birational map $\varphi : X \to \mathbb{A}^2$ given by restricting the projection $\mathbb{A}^2 \times \mathbb{P}^1 \to \mathbb{A}^2$. The map φ gives an isomorphism $X \setminus \varphi^{-1}(0) \to \mathbb{A}^2 \setminus \{0\}$, and the *exceptional curve* $E = \varphi^{-1}(0) \subset X$ is isomorphic to a $\mathbb{P}^1$ which parametrises lines in $\mathbb{A}^2$ passing through 0. Namely, the line $\ell \subset \mathbb{A}^2$ with parametric equation $(x, y) = (a_1 t, a_2 t)$ for $(a_1, a_2) \neq (0, 0)$ corresponds to the point $(0, 0, [a_1 : a_2]) \in E$ where the *strict transform* $\widetilde{\ell} = \overline{\varphi^{-1}(\ell \setminus \{0\})}$ meets E.

Compute the strict transform of a nodal cubic.

§ 2 Classification of algebraic varieties

Classification of varieties up to birational equivalence, i.e. finding a 'good representative' in every birational equivalence class, is a fundamental problem in algebraic geometry. Minimal model program (MMP)– define good, ask how representatives are related, find moduli if possible. o

§ 2.1 Curves. Every birational equivalence class of curves contains a unique smooth representative, which is projective. The *genus* (defined as $g(C) = h^0(C, \omega_C)$) is a discrete birational invariant which we can use to describe birational classes. The classification takes the following shape.

C	$\mathbb{P}^1$	elliptic curves	curves of general type
genus	0	1	$\geqslant 1$
$\deg(K_C)$	< 0	0	> 0
Kodaira dimension $\kappa(C, K_C)$	$-\infty$	0	1

This trichotomy will appear later.

§ 3 Surfaces

Inspired by curves, one might consider smooth projective surfaces as the good representatives of birational classes. This fails for the following reason.

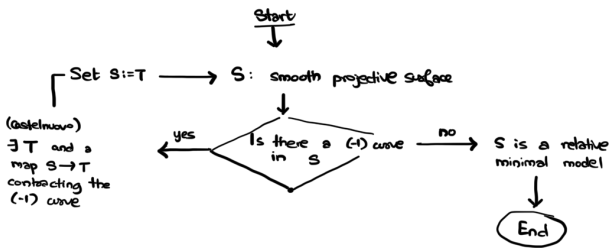
Theorem 3.1. *Let* T *be a smooth projective surface, and* $p \in T$ *a point. Then the surface* S *obtained as the blow up* $\mu : S \to T$ *of* T *in* p *has the following properties.*

1. *S is a smooth projective surface.*
2. $S \setminus \mu^{-1}(p)$ *is isomorphic to* $T \setminus p$.
3. $E = \mu^{-1}(p)$ *is isomorphic to* $\mathbb{P}^1$*, and has self–intersection* $E^2 = -1$.
4. $\mathrm{Pic}(S) \cong \mu^* \mathrm{Pic}(T) \oplus \mathbb{Z} \cdot E$

Exercise 3.2. Show that a reduced irreducible curve E inside a smooth projective surface S is a -1 curve if and only if $E^2 < 0$ and $E \cdot K_S < 0$.

Theorem 3.3 Castelnuovo's contraction theorem. *If* S *is a smooth projective surface containing a* (-1)*–curve* E*, then there exists a smooth projective surface* T *and a morphism* $\mu : S \to T$ *called the* contraction of E *such that* $\mu(E)$ *is a point* $p \in T$ *and* $S \setminus E$ *is isomorphic to* $T \setminus \{p\}$*. In fact,* μ *is the blow up of* T *in* p.

We can use this to find good representatives in a given birational class via the following algorithm.



This process terminates, since the Picard rank drops by 1 after every contraction. The outcome of this procedure can be any of the following:

1. $\mathbb{P}^2$, i.e. the del Pezzo surface of degree 9.
2. Minimal geometrically ruled surfaces S, which are $\mathbb{P}^1$–bundles over C.
3. Minimal models, which are characteristed by having non–negative Kodaira dimension.

The first two cases are characterised by having Kodaira dimension $-\infty$, and are the so-called *Mori fiber spaces*. Their birational classes may have multiple good representatives, Sarkisov program.

§ 3.1 MMP in dimension $\geqslant 3$. Given a smooth variety X, we expect a sequence of birational surgery operations which produce a (possibly mildly singular) variety Y which comes equipped with a fibration $f : Y \to Z$ (projective surjective morphism with connected fibers). We again have a dichotomy–

1. if X is uniruled (i.e. covered by rational curves), then $f : Y \to Z$ is a Mori fiber space and satisfies $\dim Z < \dim Y$ with relative Picard rank 1 and the general fiber F is a Fano variety with terminal singularities. Conjecturally this is the case of negative Kodaira dimension, and is proven for dimension $\leqslant 3$.
2. If X is not uniruled then we obtain a variety Y called a *minimal model*, which has nef canonical class. The existence of $f : Y \to Z$ is predicted by the *abundance conjecture*, which asserts that the linear system $|mK_Y|$ is non–empty and basepoint free for some $m \in \mathbb{N}$. If $F$is a general fiber of f, then the Iitaka fibration theorem says that K_F is trivial i.e. F is a Calabi Yau variety and further, Z is a canonically polarised variety i.e. K_Z is ample.

THe MMP thus predicts that there are three building blocks in birational geometry– Fano varieties, Calabi–Yau varieties, and canonically polarised varieties. For hypersurfaces in $\mathbb{P}^N$, these correspond to degrees $\leqslant N, = N+1, \geqslant N+2$ respectively.

Exercise 3.4. If C is a smooth projective curve of genus $\geqslant 2$, show that $h^0(C, mK_C) = (2m-1)(g-1)$ for $m \geqslant 2$.

Exercise 3.5. If S is a minimal smooth projective surface of Kodaira dimension 2, show that $h^0(X, mK_S) = \frac{1}{2}m(m-1)K_S^2$-

§ 4 Preliminaries

§ 4.1 Canonical class. If X is a smooth variety of dimension n, then have a cotangent bundle Ω^1_X and the canonical bundle $\omega_X = \bigwedge^n \Omega^1$. The canonical class K_X is any divisor such that $\omega_X = \mathcal{O}_X(K_X)$.

K_X makes sense even if X is just normal, by taking the canonical class of the smooth locus and extending (uniquely) by Hartog's lemma. This is a Weil divisor, well-defined up to linear equivalence.

§ 4.2 Iitaka dimension. If X is a projective variety and $L = \mathcal{O}(D)$ is a line bundle, then we wish to understand the asymptotic behaviour of $h^0(L^n)$. Base locus of L is the set of all points in X where every global section of L vanishes, then have a map

$$\varphi = \varphi_{|L|} : X \setminus \mathrm{Base}|L| \to \mathbb{P}H^0(X, L).$$

Write $\mathbb{N}(L)$ for the set of all m for which $H^0(X, L^m) \neq 0$. If $m \in \mathbb{N}(L)$ then have a rational map $\varphi_{|L^m|}$ with image Y_m.

Definition 4.1. For X, L as above and X normal, the Iitaka dimension of L is defined as

$$\kappa(X, L) = \begin{cases} \max_m(\dim Y_m), & \mathbb{N}(L) \neq \{0\} \\ -\infty, & \mathbb{N}(L) = \{0\}. \end{cases}$$

Thus $\kappa(X, L) \in \{-\infty, 0, 1, ..., \dim X\}$. If X is a smooth projective variety, then $\kappa(X) = \kappa(X, \omega_X)$ is called the Kodaira dimension of X.

Remark 4.2. If $\kappa(X, L) = k$ then $\dim \varphi_m(X) = k$ for all sufficiently large $m \in \mathbb{N}(L)$. In particular, $k = 0$ if and only if $h^0(X, L^m) \leqslant 1$ for all m and $h^0(X, L^n) = 1$ for some $n > 0$.

Remark 4.3. If $\kappa(X, L) = k$ then there exist $\alpha, \beta > 0$ such that $\alpha m^k \leqslant h^0(X, L^m) \leqslant \beta m^k$ for sufficiently large $m \in \mathbb{N}(L)$.

THe definition also makes sense for $\Bbbk$-Cartier $\Bbbk$-divisors for $\Bbbk \in \{\mathbb{Q}, \mathbb{R}\}$, by rounding down mD when taking global sections.

Remark 4.4. If $f : Y \to X$ is a surjective morphism between normal varieties and D is a $\mathbb{Q}$-cartier $\mathbb{Q}$-divisor on X then $\kappa(X, D) = \kappa(Y, f^*D)$.

If $f : Y \to X$ is a projective birational morphism between normal varieties and D is a $\mathbb{Q}$-cartier $\mathbb{Q}$-divisor on X, then for every effective f-exceptional divisor $E \subset Y$ we have $\kappa(X, D) = \kappa(Y, f^*D + E)$.

§ 4.3 Notions of positivity. Let X be a normal projective variety, and $L = \mathcal{O}(D)$ a line bundle on X. Then L is *globally generated* if $\mathrm{Base}|L| = \emptyset$, equivalently if the natural map $\mathcal{O} \otimes H^0(X, L) \to L$ is surjective, equivalently if the rational map $\varphi_{|L|}$ is a morphism.

L is semiample if there is some $m > 0$ such that L^m is globally generated.

L is ample if there is some $m > 0$ such that L^m is globally generated and $\varphi_{|L^m|}$ is an embedding.

L is nef if $\int_C c_1(L) \geqslant 0$ for every irreducible curve $C \subset X$, equivalently if $D \cdot C \geqslant 0$ for every irreducible curve $C \subset X$.

L is big if $\kappa(X, L) = \dim X$. Then D is big iff and only if there is an $m > 0$ such that $mD = A + E$ for A ample and E effective. Then $\varphi_{|mD|}$ is birational.

D is pseudoeffective if $[D] = \lim_{k\to\infty}[E_k]$ for $[E_k]$ a sequence of effective divisors in $\mathrm{NS}(X)$.

We have the following.

$$\text{ample} \Rightarrow \text{semiample} \Rightarrow \text{nef} \Rightarrow \text{pseudoeffective}, \qquad \text{ample} \Rightarrow \text{big} \Rightarrow \text{pseudoeffective}.$$

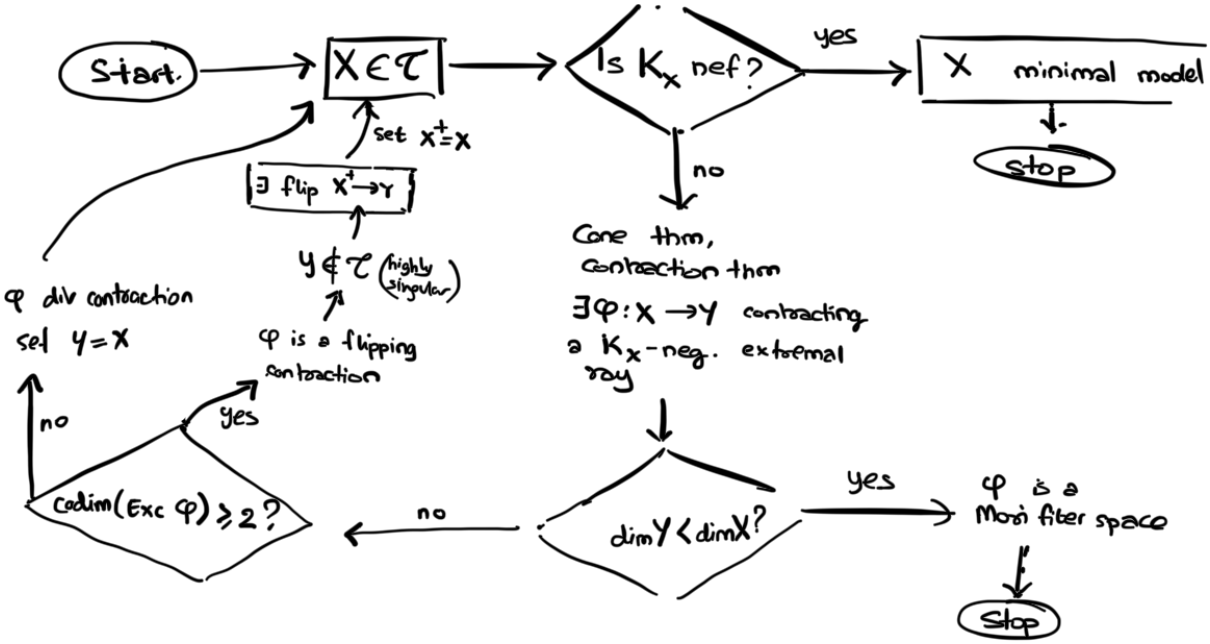
We consider the following real vector spaces- the Neron Severi space $N^1(X)_{\mathbb{R}} = \mathrm{Div}(X)/\text{numerical equivalence} \otimes \mathbb{R}$ is finite dimensional and its dimension $\rho(X)$ is the Picard number of X. $N_1(X)_{\mathbb{R}} = Z_1(X)/\equiv \otimes \mathbb{R}$ is the realification of group of numerical equivalence classes of 1-cycles.

Consider the cones $\mathrm{Amp}(X), \mathrm{Nef}(X), \mathrm{Big}(X), \overline{\mathrm{Eff}}(X)$. We know that the nef cone is the closure of the ample cone and the ample cone is the interior of the nef cone. Likewise the closure of the big cone is the pseudoeffective cone, and the interior of the pseudoeffective cone is the big cone.

Inside $N_1(X)_{\mathbb{R}}$ have $NE(X)$ generated by classes of irreducible curves, and its closure $\overline{NE}(X)$ the Kleiman–Mori cone.

Exercise 4.5. Let S be a smooth projective surface with Kodaira dimension $\geqslant 0$. Show that S contains no (-1)-curves if and only if K_S is nef.

§ 5 The MMP in arbitrary dimensions



Existence of flips has been solved in "full generality" (2019), Mori got fields medal ('90s) for proving the existence of flips of terminal 3-folds.

Question: does the procedure terminate? Whenever we have a divisorial contraction $\varphi : X \to Y$, the Picard rank drops so we can only have finitely many divisorial contractions when running the MMP. Flipping contractions preserve the Picard rank.

Conjecture (termination of flips): there can only appear finitely many flips when running the MMP.

The MMP has been established in full generality in dimension 3 in the 80s and 90s by the works of Kawamata, Mori, Ried, Shokurov, and many others.

The MMP studies pairs (objects of form X, B) which offer flexibility and are great for inductive purposes, and generalised pairs $(X, B + M)$

Pseudoeffectivity of K_X determines whether or not you get minimal model or a Mori fiber space.

§ 5.1 Cone and contraction theorems. Cone theorem: The Kleiman mori cone $\overline{NE}(X)$ is the sum of the K_X-non-negative part and the span of K_X-negative extremal rays $\sum_i \mathbb{R}_{\geqslant 0}[\gamma_i]$ where each $\gamma_i \subseteq X$ is an irreducible rational curve. This relies on characteristic p methods (Mori's bend and break technique). Further for any $\varepsilon > 0$, there are finitely many γ_i with $K_X \cdot \gamma_i > \varepsilon$. But may have accummulation near the hyperplane $K_X = 0$.

Contraction theorem: every K_X-negative extremal ray R can be contracted by a morphism $\varphi : X \to Y$, $\varphi_* \mathcal{O}_X = \mathcal{O}_Y$, and where a curve $C \subset X$ is contracted by φ if and only if $[C] \in R$.

§ 6 Pairs and their singularities

If X is a smooth variety, a reduced $\mathbb{Q}$-divisor $D = \sum D_i$ is SNCif each D_i is smooth and all components intersect transversally. If not reduced then we say it is SNC when the underlying reduced divisor is so.

A resolution of X is a projective birational morphism $g : Y \to X$ where Y is smooth.

X a variety, D a $\mathbb{Q}$-divisor on X. A log-resolution of (X, D) is a projective birational morphism $g : Y \to X$ from a smooth variety Y such that the exceptional locus of g is a divisor and $\mathrm{Exc}(g) \cup \tilde{D}$ has SNC support.

Theorem (Hironaka) Log resolutions exist over fields of characteristic 0.

X a variety, $f : Y \to X$ a projetive birational morphism from Y normal. $E \subset Y$ prime divisor. Then we say E is a divisor over X and $f(E)$ is called the center of E on X, written $c_X(E)$. Divisorial valuations of $k(X)$?

Definition 6.1. A pair (X, B) consists of a normal variety X together with an effective $\mathbb{Q}$-divisor B on X such that $K_X + B$ is $\mathbb{Q}$-cartier.

Definition 6.2. If X is a normal variety and B is a $\mathbb{Q}$-divisor on X such that $K_X + B$ is $\mathbb{Q}$-cartier. If $f : Y \to X$ is a resolution, pick canonical divisors K_Y, K_X such that $f_* K_Y = K_X$. Then $K_Y = f^*(K_X + B) + \sum a(E_i, x, B)E_i$. The rational number $a(E_i, x, B)$ is called the discrepancy of E_i with respect to (X, B). Note $f^{-1}B$ always has discrepancy 1.

The discrepancy of (X, B) is defined as the infimum of all discrepancies $a(E, X, B)$ where Eis a prime divisor over X which is exceptional.

The total discrepancy of (X, B) is the defined as the infimum of all discrepancies $a(E, X, B)$ for all prime divisors E over X.

Every divisor on X other than B has discrepancy zero by definition.

Exercise 6.3. Let X be a smooth variety and let $B = \sum a_i D_i$ be a $\mathbb{Q}$-divisor on X. Let $Z \subset X$ be a closed subvariety of codimension n, and let $p : Y \to X$ be the blow up of X along Z and $E \subset Y$ the component of exceptional divisor dominating Z. Show that $a(E, X, B) = r - 1 - \sum a_i \text{mult}_Z(D_i)$.

Exercise 6.4. Let $f : Y \to X$ bea proper birational morphism between normal varieties. Let B_X resp B_Y be a $\mathbb{Q}$-divisor on X resp Y such that $K_Y + B_Y = f^*(K_X + B_X)$ and $f_* B_Y = B_X$. Show that for any divisor F/X we have $a(F, X, B_X) = a(F, Y, B_Y)$.

Lemma 6.5. *If* (X, B) *is as before, then either* $\text{discrep}(X, B) = -\infty$ *or* $-1 \leqslant \text{totaldiscrep}(X, B) \leqslant \text{discrep}(X, B) \leqslant 1$.

If $(X, B = \sum a_i D_i)$ *is a log-smooth pair (ie* X *smooth and* B *has SNC support) then* $\text{discrep}(X, B) = \min\{1, \min\{1 - a_i\}, \min_D$

Definition 6.6 (Singularities of pairs). We say a pair (X, B) is terminal if $\text{discrep}(X, B) > 0$, canonical if $\text{discrep}(X, B) \geqslant 0$, klt if $\text{discrep}(X, B) > -1$ and $\lfloor B \rfloor = 0$, and log canonical if $\text{discrep}(X, B) \geqslant -1$.

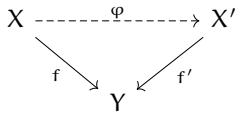
Proposition 6.7. *Let* X *be a normal variety such that* K_X *is* $\mathbb{Q}$*-cartier. Let* $f : Y \to X$ *be a resolution. Write* $K_Y = f^* K_X + \sum a_i E_i$. *Then* X *is canonical if and only if* $a_i \geqslant 0$ *for all* i*, and* X *is terminal if and only if* $a_i > 0$ *for all* i.

Proposition 6.8. *Let* (X, B) *be a pair,* B *not necessarily effective. Let* $f : Y \to X$ *be a log resolution. Suppose* $K_Y + f^{-1}B = f^*(K_X + B) + \sum a_i E_i$. *Then* (X, B) *is klt if and only if* $a_i > -1$*, and log canonical iff* $a_i \geqslant -1$.

Exercise 6.9. Let $X = \mathbb{A}^2$ and $B = \text{div}(x^2 - y^3)$. The pair (X, dB) is KLT (resp. log canonical) iff $0 \leqslant d < 5/6$ (resp. $0 \leqslant d \leqslant 5/6$).

Lemma 6.10 Negativity lemma. *Let* $h : Z \to Y$ *be a proper birational morphism. Let* D *be a* $\mathbb{Q}$*-cartier* $\mathbb{Q}$*-divisor on* Z *such that* $-D$ *is* h*-nef. Then* $D \geqslant 0$ *if and only if* $h_* D \geqslant 0$.

Exercise 6.11. Consider a commutative diagram



for X, X' normal varieties and f, f' projective birational. Let B, B' be $\mathbb{Q}$-divisors on X, X' respectively. Assume $f_* B = f'_* B'$, $-(K_X + B)$ is Q-cartier and f-nef, $K_X + B'$ is Q-cartier and f'-nef. Show that every exceptional divisor E/Y we have $a(E, X, B) \leqslant a(E, X', B')$ and show that the strict inequality holds if either $-(K_X + B)$ is f-ample and f is not an isomorphism above generic point of $c_Y(E)$, or $K_{X'} + B'$ is f'-ample and f' is not an iso above geneirc of cY(E).

This models divisorial contractions and flips, so MMP doesnt make singularities worse.

Definition 6.12. Let X, B be a pair. A $(K_X + B)$-flipping contraction is a proper birational morphism $\theta : X \to W$ such that the codimension of exceptional locus is $\geqslant 2$ and $-K_X + B$ is θ-ample.

A pair (X^+, B^+) together with a proper birational morphism $\theta^+ : X^+ \to W$ is called a $(K_X + B)$-flip of θ if codim of exceptional θ^+ is $\geqslant 2$ and $K_{X^+} + B^+$ is θ^+-ample.

Remark 6.13. If a variety X is terminal then it is smooth in codimension 2 i.e. the singular locus has codimension $\geqslant 3$. Thus terminal surfaces are smooth. Further if (X, B) is terminal then $\lfloor B \rfloor = 0$.

Definition 6.14. Let $(X, B = \sum a_i D_i)$ be a pair and $0 \leqslant a_i \leqslant 1$ for all i. Then (X, B) is dlt (divisorially log terminal) if there is a closed subset $Z \subset X$ such that

1. $(X \setminus Z, B|_{X\setminus Z})$ is log-smooth
2. If $f : Y \to X$ is a birational morphism and $E \subset Y$ is a prime divisor with $c_X(E) \subset Z$ then $a(E, X, B) > -1$.

Thus dlt implies log canonical, and dlt is klt iff $\lfloor B \rfloor = 0$. DLT is KLT away from log-smooth locus.

§7 Flips and termination

Definition 7.1. For (X, B) a pair, a $K_X + B$-flipping contraction is a proper birational morphism $\theta : X \to W$ such that $\operatorname{codim}(\operatorname{Exc}(\theta)) \geqslant 2$ and $-(K_X + B)$ is θ-ample. In this case a pair (X^+, B^+) with a proper birational morphism $\theta^+ : X^+ \to W$ is called a $(K_X + B)$-flip if $\operatorname{codim}\operatorname{Exc}(\theta^+) \geqslant 2$ and $K_{X^+} + B^+$ is θ^+-ample.

In this case B^+ is the strict transform of B.

If a flip exists then it has to be unique, and $X^+ = \operatorname{Proj}_W \bigoplus_{m\geqslant 0} \theta_* \mathcal{O}_X(m(K_X + B))$. The existence of flips is equivalent to the ring being finitely generated, and the question has been settled in full generality (Mori '88 showed this for terminal 3-folds, Shokurov '92 proved the existence for KLT pairs, Kollar et al (book Geometry of algebraic 3-folds) showed existence of log canonical pairs in dimension 3, BCHM '10 showed the existence of flips for KLT pairs of any dimension, Birkar '12 and Hacon-Xu '13 showed existence of flips for LC pairs in any dimension. All of this for relative Picard rank 1, general by ???)

Lemma 7.2 (monotonicity). *Let* $(X, B) \dashrightarrow (X^+, B^+)$ *be a* $(K_X + B)$ *flip over* W. *For any divisorial valuation* F/X *we have* $a(F, X, B) \leqslant a(F, X^+, B^+)$ *and the strict inequality holds if and only if either* $c_X(F) \subseteq \mathit{Exc}(\theta)$ *or* $c_{X^+}(F) \subset \mathit{Exc}(\theta^+)$.

Lemma 7.3. *have* $\dim(\mathit{Exc}\theta) + \dim(\mathit{Exc}\theta^+) \geqslant \dim X - 1$. *Thus in dimension* 3, *the only possibility is that a bunch of curves get flipped to a bunch of curves. In dimension 4, all combinations* $(2,1), (1,2), (2,2)$ *are possible.*

Lemma 7.4. *For* (X, B) *a log canonical pair, if* X *is* $\mathbb{Q}$*-factorial then so is* X^+ *and further, the Picard rank is preserved.*

(STATES THE TERMINATION OF FLIPS CONJECTURE)

Shokurov '85 proved that the conjecture holds for terminal 3-folds with no boundary. Kawamata '92 showed that the conjecture holds for $\mathbb{Q}$-factorial dlt 3-folds. Shokurov '96 showed that conjecture holds for LC pairs in dimension 3. Kawamata–Matsuda–Matsuki '87 showed conjecture holds for terminal 4-folds (no boundary). Fujino '04–'05 showed that conjecture holds for canonical pairs of dimension 4. Alexeev–Hacon–Kawamata '07 showed conjecture holds for klt pairs of dimension 4 with $-K_X - B \equiv E \geqslant 0$. Birkar '10 showed that the conjecture holds for KLT pairs of dimension 4 with Kodaira dimension of $K_X + B$ at least 2. Hacon–Moraga '20, Chen–Tsakanikas '23 show that the conjecture holds for LC pairs (X, B) of dimension 4 such that $K_X + B$ is pseudoeffective. Lenn–Pacienza '10 show that any MMP starting from an IHS pair (i.e. log canonical pair with underlying smooth IHS variety) terminates. Denisi–Ortiz–Tsakanikas–Xie '24 show that any MMP starting from an Enriques pair (log canonical pair with underlying smooth Enriques manifold) terminates.

§7.1 Shokurov's proof. He defines a special invariant.

Definition 7.5. If X is a normal projective with terminal singularities, then the difficulty of X is the number of E exceptional over X such that $a(E, X, 0) < 1$.

He shows this is well-defined (finite) and upon flipping, it does not increase.

Thus it suffices to show that the difficulty decreases after a flip. This is true in dimension 3. We know X^+ is terminal, so X^+ is smooth at the generic point of every flipped curve. Pick such a C^+ and blow up X^+ at the generic point of this curve, and denote by E^+ the exceptional divisor dominating C^+. Then [Exercise] show that $a(E^+, X, 0) < a(E^+, X^+, 0)$ since center of E^+ lies in the flipped locus. But $a(E^+, X^+, 0) = 1$, so the difficulty of X is bigger than that of X^+.

In dimension 4, if the flipping and flipped loci have dimensions $(1,2)$ or $(2,2)$ then again difficulty decreases. But $(2,1)$ needs a different method.

§7.2 Very rough sketch of Chen–Tsakanikas. it is a special case of the folowing more general statement:

Theorem 7.6. *Assume termination of flips for klt generalised pairs of dimension* $\leqslant n-1$. *Let* $(X, B+M)$ *be a log canonical generalised pair of dimension n such that* K_X+B+M *is pseudoeffective. If* $(X, B+M)$ *has a minimal model then any sequence of flips starting from* $(X, B+M)$ *terminates. (If some MMP terminates then all MMPs terminate).*

Main ideas are special termination and ascending chain condition for log canonical thresholds proved by (Hacon–McKernan–Xu).